

ORIGINAL ARTICLE

Optimal design and analysis of a grid-connected hybrid renewable energy system using HOMER Pro: A case study of Tumbatu Island, Zanzibar

Thani R. Said¹  | Baraka Kichonge^{1,2}  | Thomas Kivevele¹ 

¹School of Materials Energy Water and Environmental Sciences (MEWES), The Nelson Mandela African Institution of Science and Technology (NM-AIST), Arusha, Tanzania

²Mechanical Engineering Department, Arusha Technical College, Arusha, Tanzania

Correspondence

Thomas Kivevele, School of Materials Energy Water and Environmental Sciences (MEWES), The Nelson Mandela African Institution of Science and Technology (NM-AIST), Tanzania, Arusha, Tanzania.
Email: thomas.kivevele@nm-aist.ac.tz

Abstract

This study addresses the pressing issue of quality electricity access in remote regions, with a specific focus on Tumbatu Island in Tanzania. Most studies on quality improvement concentrate on low-voltage distribution lines and leave high-voltage (HV) transmission lines behind. It is essential to address quality issues in HV lines due to their critical role in electricity transmission and distribution infrastructure. The objective of the study is to improve the voltage profile on the island's HV transmission line by identifying the optimal hybrid energy system comprising solar PV, wind turbine, and battery technologies. The study begins by presenting the total power demand and consumption on Tumbatu Island, which are important factors in designing an efficient energy system. The findings reveal a power demand of 7173 kW and a consumption of 28540 kWh/day, with an average scaled value of 1507.9 and 6000 kWh/day. To achieve the desired voltage profile improvement, the research incorporates HOMER Pro software for simulating and analyzing various hybrid system configurations. Essential input parameters, such as costs, resources, technology components, and load, are considered in the simulation process. The software generates a ranked list of options based on the system's net present cost (NPC). The simulation results demonstrate that the integration of solar PV, wind turbine, and HV lines provides a substantial improvement in the voltage profile, increasing it from 29.6 to 31.23 kV during peak demand periods. This solution proves to be the most economically viable, with the lowest NPC of \$4,003,851 and a relatively short payback period of 3.79 years. In addition, a sensitivity analysis is performed to identify the most influential parameter in the system's performance. Wind speed is found to have the greatest impact, emphasizing its significance in the design and operation of the hybrid energy system. The implementation of this optimal hybrid renewable energy system on Tumbatu Island will not only improve the voltage profile and meet the island's energy needs but also shall contribute to

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global efforts in reducing pollution and the cost of electricity to the Tumbatu Island population. Moreover, it addresses the current and future demands for clean energy, underlining its importance in achieving sustainable and accessible electricity in sub-Saharan Africa and beyond.

KEYWORDS

clean energy, high voltage, HOMER, solar PV, voltage profile, wind turbine

1 | INTRODUCTION

Energy is critical for economic progress, and its demand is increasing on a daily basis because of population growth, urbanization, and rapid industrialization.^{1,2} However, increases in demand offset gains made through resource usage efficiency improvements and the expansion of renewable energy production. To meet the world's rising energy demand, fossil fuels have come to dominate the overall energy consumption profile, accounting for up to 80% of total energy consumption.³ However, using fossil fuels to meet global energy demands comes with a number of socioeconomic and environmental concerns. Fossil fuels emit significant amounts of carbon dioxide^{4,5} and are vulnerable to resource depletion,^{6,7} therefore raising concerns about environmental safety and resource sustainability.⁸ To address these concerns, countries need to adjust their energy mix by reducing the reliance on fossil fuels and increasing the use of renewable energies,⁹ seen as the clean energy for the future.¹⁰ However, renewable energies are faced by the major challenges of energy production due to its intermittent nature in different climatic conditions.^{11,12} Intermittent renewable energy sources (RES) are challenging in implementation because they disrupt conventional methods of planning the daily operation of the electric grid. Electrical energy from renewable sources fluctuates along several time horizons, necessitating the grid operator to adjust its day-ahead, hour-ahead, and real-time operational procedures.¹³ Such a drawback of RES can be overcome by combining more than one type of energy into a grid to increase electricity output efficiency.¹⁴ Islands such as Tumbatu in Zanzibar have a wealth of renewable energy resources, and by utilizing a variety of renewable energy technologies, small islands can develop by satisfying all of their domestic energy needs.¹⁵ Tumbatu Island is the third-largest in the Zanzibar archipelago, covering an area of 14 square km, located to the Northwest of Unguja Island. According to National Bureau of Statistics 2022, Tumbatu has a population of 26,482, with average house hold of 4.5%. A transmission line that runs from Mtoni substations in Zanzibar to Mkokotoni around 35.3 km supplies electricity

to the island of Tumbatu via submarine cable (33 kV). The transmission line has a number of challenges, the most serious of which is voltage drop to 30 kV due to the long distance (35.3 km) and lack of enough power to fulfill the current and growing power demand.

In the quest to find solutions for the ongoing voltage drop issue, one possible course of action is to construct a new overhead transmission line that is connected by a submarine cable. However, a critical issue that obstructs the feasibility of this approach is its financial viability, primarily attributed to the substantial costs involved. The installation of a new submarine cable comes with considerable expenses,^{16,17} further compounded by its considerable distance from the Zanzibar substation. Moreover, there is a potential environmental concern related to maritime contamination, adding another layer to the non-viability of the proposed project.^{14,18} Given these challenges, it becomes evident that the remaining feasible socially and economically available solution lies in the utilization of renewable energy integrated with grid electricity.^{19–21} By embracing RES, such as solar or wind power, and integrating them with the existing grid infrastructure, it is possible to address the voltage drop issue more sustainably and economically.²² Not only does the utilization of renewable energy promote environmental sustainability,²³ but it also offers long-term cost savings.^{24,25} By reducing reliance on traditional energy sources and harnessing the power of renewable resources, the electricity grid becomes more resilient and less prone to voltage drop issues.^{26,27} This approach aligns with the global shift toward a greener and more sustainable energy future. Therefore, it is imperative to shift the focus toward exploring and implementing solutions that incorporate renewable energy into the existing grid infrastructure. This approach not only addresses the voltage drop issue but also contributes to the overall sustainability goals and energy independence of the region.

Numerous studies have thoroughly investigated the integration of photovoltaic (PV) systems into low-voltage (LV) networks to address challenges such as voltage drop. For instance, Mouheb et al.²⁸ utilized Jp élec

software to analyze data, revealing a 10% reduction in voltage drop and improved electrical quality when connecting PV systems during peak demand. In a similar vein, Adel and Rachid²⁹ explored connecting a hybrid PV-wind turbine mini-power station to a rural LV network using PJ-elec software. Their simulation demonstrated substantial delivery of clean electricity and a noticeable enhancement in voltage (less than 10%) observed between 6 and 17 h. Moreover, Serem et al.³⁰ conducted a study on voltage profile and sensitivity analysis. Their findings suggested that integrating PV systems impacted grid stability, with multiple renewable sources slightly enhancing voltage levels. In a comparative study, Ali et al.²² favored double-fed induction generator (DFIG) over squirrel cage induction generator (SCIG) with renewable sources, emphasizing DFIG's superiority for stability. Additionally, higher solar radiation increased solar generator output, contributing to an improved electrical system voltage profile. Further insights from Kumar,³¹ MATLAB/SIMULINK study on Voltage Profile Improvement emphasized better results with solar PV than wind, consolidating the argument for the effectiveness of solar energy integration.

Exploring dynamic voltage stability, Yaghoobi et al.³² proposed D-STATCOM to compensate for PV power loss due to cloud shading in distribution systems with high PV penetration. Additionally, Pokhrel et al.³³ utilized Ant Colony Optimization (ACO) to find optimal distributed generator (DG) locations, significantly enhancing the Voltage Profile of an 11 KV system from 0.828 to 0.982 pu. In a parallel effort, Adebantia et al.³⁴ employed Static Var Compensators to reduce power loss and enhance Voltage Profile in an Electrical Power Distribution Network, achieving an impressive increase in network capacity from 150% to 263% in NEPLAN software. Particularly, Ntombela,³⁵ contributed to the discourse by minimizing power loss and improving voltage profiles in transmission networks through a network modification algorithm. Their study achieved a significant decrease in real power loss and up to 1.01 pu for the highest bus voltage in the IEEE-30 bus. In line with these compelling findings, adopting an on-grid system, as suggested by Sharma et al.,³⁶ emerges as a strategic. Unlike previous studies that primarily focused on utilizing diverse software tools like Jpelec, PJ-elec, Power World Simulator, D-STATCOM, and MATLAB/SIMULINK, this research stands out by exclusively employing the HOMER Pro software.

Most studies on quality improvement concentrate on LV distribution lines and leave high-voltage transmission lines behind. However, it is essential to address quality issues in high-voltage (HV) lines due to their critical role in electricity transmission and distribution infrastructure. HV lines are

responsible for transmitting large amounts of electricity over long distances, serving as the backbone of power distribution networks,³⁷ making them vital for powering industries, homes, and essential services.³⁸ Neglecting quality issues in HV lines can lead to significant risks such as power outages due to overloading of transmission line, equipment failures, and even safety hazards for both workers and the public.^{39,40} Additionally, addressing quality issues in HV lines can improve system reliability, reduce energy losses, enhance overall grid efficiency, and promote sustainable energy practices.⁴¹ Therefore, prioritizing quality improvement efforts in HV lines is essential for ensuring the reliability, stability, and security of the entire electrical transmission system and country as well.

In contrast to its predecessors, this study approach extends beyond the improvement of LV transmission line voltage profiles. HOMER Pro is utilized comprehensively for designing hybrid renewable energy systems (HRES), analyzing voltage profiles on HV lines, and conducting economic assessments.^{42,43} This distinctive application of HOMER Pro marks this study as a pioneering initiative, significantly advancing technical and economic performance in electric power supply, particularly in challenging topography like Tumbatu Island. To the best of the author's knowledge, no comprehensive studies on enhancing the voltage profile of HV lines in underdeveloped countries have been conducted. Therefore, the aim of this study is to evaluate the effectiveness of integrating optimized HRESs using HOMER Pro in improving the voltage profile in long HV transmission lines. The research will concentrate on Tumbatu Island in Zanzibar as a specific case study. The main objective of this investigation is to analyze the cost factors and economic implications of the project in detail, with the ultimate goal of conducting a comprehensive financial examination. Additionally, alternative approaches will be explored to comprehensively assess the project's financial feasibility, considering cost-effectiveness as a significant aspect. The article is structured into three sections: the first covering the simulation methodology using HOMER Pro, the second discussing the analysis of voltage profiles based on simulation results, and the final section exploring economic and sensitivity analyses of the optimal solution derived from the software outcomes.

2 | MATERIALS AND METHODS

2.1 | Materials

The HOMER Pro micro grid analysis tool was utilized to assess the most economically viable option of hybrid

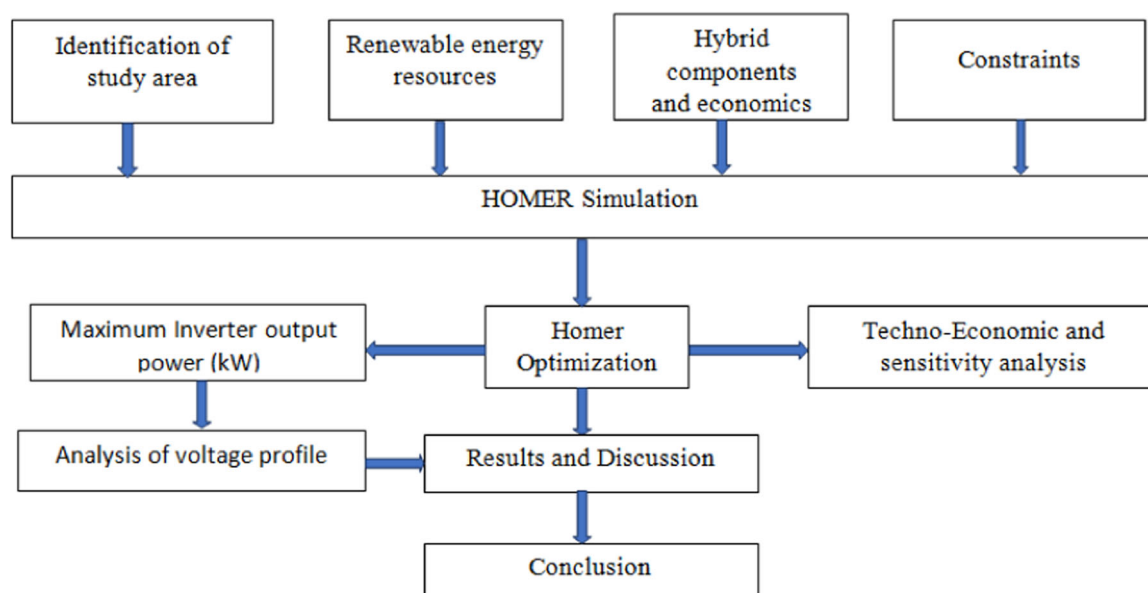


FIGURE 1 Summary of the methodology.

energy sources in this study. It simulates, optimizes, and sensitizes various parameters inputted during the design process. The software requires a computer for execution. Specifically, HOMER Pro version 3.14.2 was employed in conjunction with an HP Intel Laptop computer equipped with an 8 GB processor and 64-bit architecture.

2.2 | Methods

The hybrid energy system combines various renewable resource components, including solar PV, wind turbines, inverters, batteries, and HV transmission lines. The purpose of this study is to enhance the voltage profile in the HV line at Tumbatu Island in Zanzibar, serving as the case study area. The criteria for selecting the optimal hybrid energy combination for the proposed case study include cost, sustainability, technology maturity, and minimum cost of energy (COE). The complete methodology is explained below in steps and summarized in Figure 1.

Step 1: Designing Hybrid Optimization Model using HOMER Pro.

Step 2: Perform simulation and optimization using HOMER Pro.

Step 3: Analysis of voltage profile using optimized results from HOMER Pro.

Step 4: Perform Techno-economic and Sensitivity analysis using HOMER Pro.

Step 5: Results, Discussion and Conclusion of the work.

TABLE 1 Monthly existing voltage per phase per day.

Month	High voltage side (kV)	Low voltage side (kV)
January	29.7	0.219
February	29.7	0.219
March	30	0.22
April	30	0.22
May	30	0.22
June	30	0.22
July	30	0.22
August	30	0.22
September	30	0.22
October	30	0.22
November	29.6	0.218
December	29.6	0.218

2.3 | Existing voltage profile

Tumbatu Island is fed by a 200 kVA, 33/0.415 kV step-down transformer, with monthly maximum recorded voltages in HV and LV sides, as per local supply electricity cooperation depicted in Table 1.

2.4 | Description of HOMER

Homer is a Hybrid Optimization of multiple energy resources developed by the National Renewable Energy

Laboratory. It is one of the most effective simulation software for designing microgrids and distributed energy systems and is one of the foremost broadly used programs.⁴⁴ HOMER Pro is capable of simulating multiple system configurations using inputs such as various technological possibilities, component costs, resource availability, manufacturer's data, and so on, and generates results as a list of possible configurations ranked by net present cost (NPC). It is capable of determining load-serve policies that use the least expensive source to meet the load. Homer performs three primary functions include simulation, optimization, and sensitivity analysis.⁴⁵ In HOMER Pro, there is a window that displays the results of variation of the output AC load from the Inverter output. HOMER is structured into three key phases: the pre-HOMER process, HOMER analysis, and post-HOMER Pro analysis. The pre-HOMER process involves identifying the study area, renewable energy resources, hybrid components, and considering economic factors and constraints. The HOMER analysis stage comprises simulation, optimization, techno-economic evaluation, and sensitivity analysis. Subsequently, the post-HOMER analysis phase involves presenting the obtained results. Optimization of solar PV/wind relies on factors such as solar irradiance, wind speed, wind velocity, and a cost analysis of the various system components.⁴⁶

2.4.1 | Designing HRES model using HOMER Pro

In this section, the HOMER Pro software was utilized to design a hybrid energy system, incorporating HV line

from the utility grid, solar cells, a wind turbine, a battery bank, a bidirectional converter, and DC and AC buses, as illustrated in Figure 2. The geographical location of Tumbatu Island is depicted in Figure 3, while Figures 4 and 5 show the solar radiation, clearness index, and wind data for Tumbatu Island. Solar radiation data has been obtained from the Tanzania Meteorological Authority (TMA) whereas wind data were downloaded from the National Aeronautics and Space Administration (NASA) within the HOMER software.

2.4.2 | Electrical load input

Electricity consumption data from Zanzibar Electricity Corporation (ZECO) for the years 2016–2021 were collected for analysis. Using a linear forecast function in Microsoft Excel, the data was projected into the future and the detailed results can be seen in Table 2. Given that the load on Tumbatu Island is primarily residential, the daily load demand can be categorized into three parts: low, medium, and high consumption. The low consumption represents approximately 2%, 3%, and 4% of the total daily demand, the medium consumption covers around 5%–6% of the total daily demand, and the high consumption constitutes approximately 10%, 13%, and 15% of the total daily demand. These percentages are based on the findings proposed by Alayan⁴⁷ and Kasheem and Arefin.⁴⁸ In the context of a power system, a load profile or load curve is a graphical depiction that illustrates the variations in electrical demand or load throughout a designated period.⁴⁹ The load profile for Tumbatu Island is found in Figure 6.

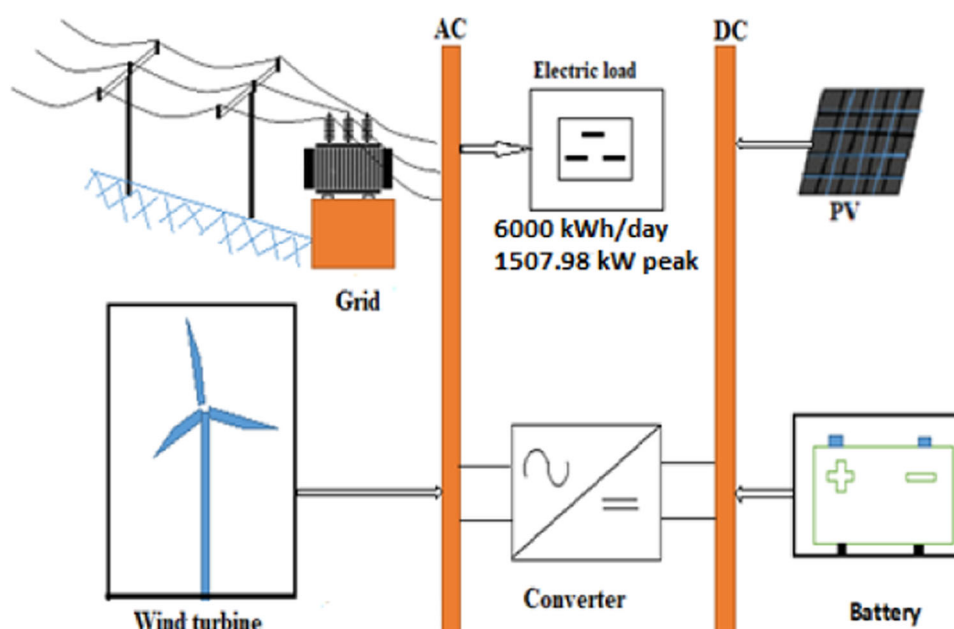


FIGURE 2 Schematic design of designed hybrid renewable energy system.



FIGURE 3 Location of Tumbatu Island (source, Google Earth).

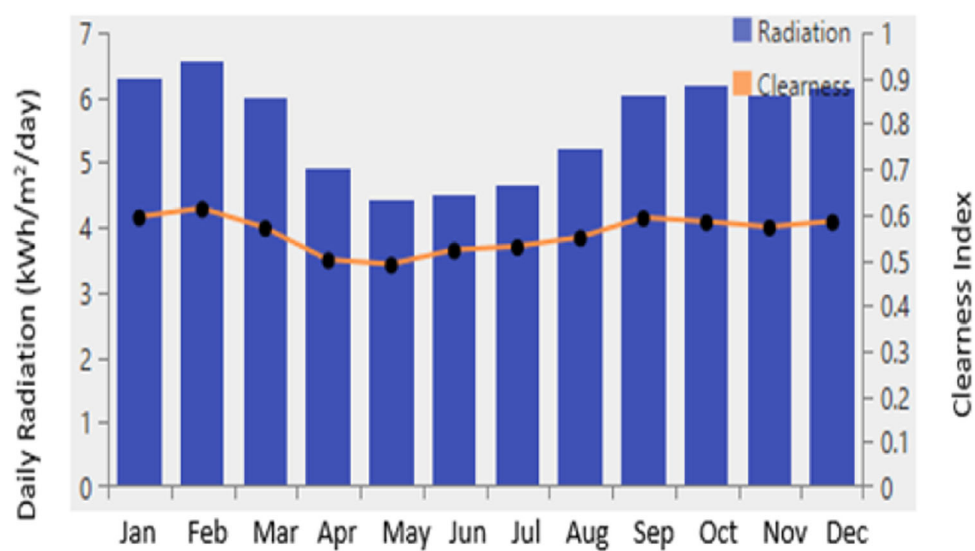


FIGURE 4 Average solar radiation (Source, TMA).

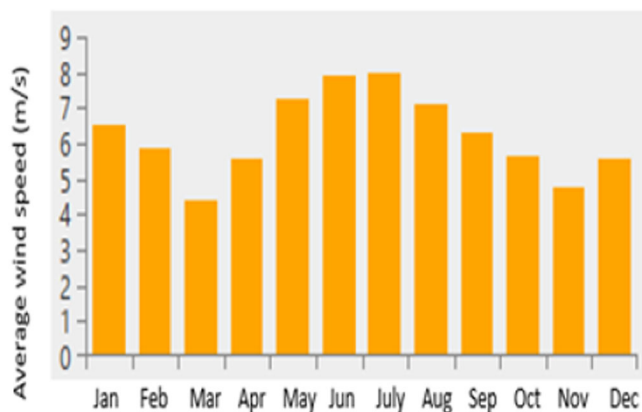


FIGURE 5 Average wind speed (Source, NASA Database).

2.4.3 | Constraints

The minimum renewable fraction of 40% was set to ensure a significant proportion of the energy generated comes from renewable sources, aligning with the goal of promoting sustainability and reducing greenhouse gas emissions. By mandating a substantial portion of renewable energy in the energy mix, the study aims to drive the transition toward cleaner and more environmentally friendly power generation. Simultaneously, the maximum annual capacity shortage of 10% was imposed to enhance grid reliability and stability. By limiting the capacity shortage to this threshold, the study seeks to mitigate the risk of power shortages and disruptions, thus safeguarding the energy supply and avoiding adverse impacts on consumers, industries, and critical services.

2.4.4 | Economics of the proposed hybrid system

The economics of the proposed hybrid system include operational, maintenance, capital, and the replacement cost value of solar panels, wind turbines, batteries, bidirectional converters, grid, and electrical load. Several factors, like labor costs, land costs, etc., have been considered in the designing of a complete HRES. The following subsections discuss in detail the economics of the hybrid components, which are summarized in Table 3.

2.4.5 | Size and cost of solar PV

A generic solar PV flat-plate with an efficiency of 17.3% has been selected to use solar energy. Their inexpensive delivery and excellent output efficiency played a

significant role in the choice to incorporate these goods into the design. The most recent global estimated PV cost is \$883 per kW.⁵⁰ The replacement cost was set at 60% of the PV price after a 25-year service life in this study, as proposed by Manyama.⁵¹ As suggested by Xu et al.,⁵² and annual operation and maintenance cost of 1% per year was also adopted.

2.4.6 | Size and cost of the wind turbine

A Generic wind turbine with rated capacity of 3 kW has been selected to convert wind energy to electrical energy for the proposed hybrid energy system as proposed in Topsham and McMillan.⁵³ The power curve generated from HOMER Pro software for the proposed wind turbine is presented in Figure 7. The most recent global estimated cost is around 1355 \$ per kW of wind turbine.⁵⁰ The operation and maintenance costs of a wind turbine has been estimated to be about 2% of its initial capital cost as adopted from Duman et al.⁵⁴ and Nsafon et al.,⁵⁵ while replacement cost is taken as 70% of capital cost as proposed in Smith et al.⁵⁶

2.4.7 | Cost and size of batteries

To determine the suitable size of the battery bank for the Hybrid Energy System, a thorough analysis is necessary. This analysis involves a detailed assessment of the battery's charging and discharging modes, considering factors like the load profile and the output energy from the RES.⁴⁶ According to adoption from Kashem and Arefine,⁴⁸ the energy from the PV panel is stored in this study utilizing a Surrette S-260 storage device, which can offer energy retained at a nominal voltage of 12 V. The selected battery has a capacity of 3.123 kWh, 83.4 Ah with maximum charge current of 16.7 A and minimum discharge current of 24.3 A. According to Mongird et al.⁵⁷ lithium-ion batteries costed \$271/kWh in 2018 and shall vary to \$189/kWh in 2025. Thus the price below 271\$/kWh and above 189 \$/kWh will be in the reasonable range for the cost of storage battery. In this study, the Installation cost of the battery is taken 256 \$/kWh. For 3.123 kWh battery, the cost per battery will be around 800 \$/battery, while replacement cost is 560 \$/battery and O & M cost is 8 \$/battery.

2.4.8 | Size and cost of power converters

A bidirectional converter has been adopted to convert DC power to AC power or vice versa because the solar PV

TABLE 2 Tumbatu hourly load demand forecast (kW) for 2030.

Month	January	February	March	April	May	June	July	August	September	October	November	December
Daily load (kWh)												
Variation of hourly electricity consumption (%)												
Hours	27,499	28,202	27,903	24,036	26,949	27,248	27,015	28,395	28,687	27,786	28,617	30,145
0	549.98	564.04	558.06	480.72	538.98	544.96	540.3	567.9	573.74	555.72	572.34	602.9
1	549.98	564.04	558.06	480.72	538.98	544.96	540.3	567.9	573.74	555.72	572.34	602.9
2	549.98	564.04	558.06	480.72	538.98	544.96	540.3	567.9	573.74	555.72	572.34	602.9
3	549.98	564.04	558.06	480.72	538.98	544.96	540.3	567.9	573.74	555.72	572.34	602.9
4	549.98	564.04	558.06	480.72	538.98	544.96	540.3	567.9	573.74	555.72	572.34	602.9
5	549.98	564.04	558.06	480.72	538.98	544.96	540.3	567.9	573.74	555.72	572.34	602.9
6	824.97	846.06	837.09	721.08	808.47	817.44	810.45	851.85	860.61	833.58	858.51	904.35
7	1374.95	1410.1	1395.15	1201.8	1347.45	1362.4	1350.75	1419.75	1434.35	1389.3	1430.85	1507.25
8	549.98	564.04	558.06	480.72	538.98	544.96	540.3	567.9	573.74	555.72	572.34	602.9
9	549.98	564.04	558.06	480.72	538.98	544.96	540.3	567.9	573.74	555.72	572.34	602.9
10	824.97	846.06	837.09	721.08	808.47	817.44	810.45	851.85	860.61	833.58	858.51	904.35
11	824.97	846.06	837.09	721.08	808.47	817.44	810.45	851.85	860.61	833.58	858.51	904.35
12	1649.94	1692.12	1674.18	1442.16	1616.94	1634.88	1620.9	1703.7	1721.22	1667.16	1717.02	1808.7
13	549.98	564.04	558.06	480.72	538.98	544.96	540.3	567.9	573.74	555.72	572.34	602.9
14	549.98	564.04	558.06	480.72	538.98	544.96	540.3	567.9	573.74	555.72	572.34	602.9
15	549.98	564.04	558.06	480.72	538.98	544.96	540.3	567.9	573.74	555.72	572.34	602.9
16	549.98	564.04	558.06	480.72	538.98	544.96	540.3	567.9	573.74	555.72	572.34	602.9
17	1099.96	1128.08	1116.12	961.44	1077.96	1089.92	1080.6	1135.8	1147.48	1111.44	1144.68	1205.8
18	1649.94	1692.12	1674.18	1442.16	1616.94	1634.88	1620.9	1703.7	1721.22	1667.16	1717.02	1808.7
19	2749.9	2820.2	2790.3	2403.6	2694.9	2724.8	2701.5	2839.5	2868.7	2778.6	2861.7	3014.5
20	3574.87	3666.26	3627.39	3124.68	3503.37	3542.24	3511.95	3691.35	3729.31	3612.18	3720.21	3918.85
21	4124.85	4230.3	4185.45	3605.4	4042.35	4087.2	4052.25	4259.25	4303.05	4167.9	4292.55	4521.75
22	1374.95	1410.1	1395.15	1201.8	1347.45	1362.4	1350.75	1419.75	1434.35	1389.3	1430.85	1507.25
23	824.97	846.06	837.09	721.08	808.47	817.44	810.45	851.85	860.61	833.58	858.51	904.35

and battery bank generate DC power while the load operates in AC mode.

In spite of the load being sourced from both the grid and RES, the inverter's rated power would be installed below, equal to, or exceeding the peak load. The current study takes into account a 1 kW bidirectional converter with a 95% efficiency as adopted from Das.⁵⁸ The converter's capital cost is \$800, the replacement cost is \$560, or 70% of the capital cost, O& M cost is taken as \$8 as proposed in Manyama.⁵¹

2.4.9 | Cost of electricity (COE)

In this study, the grid plays a vital role as an electrical backup and acts as an absorber when electricity is generated by renewable sources. The main objective of utilizing the grid as an electrical backup is to ensure a dependable and consistent supply of electricity, especially when the quantity of electricity produced from renewable sources is insufficient to meet the demand of consumers. This insufficiency can be caused by various factors, such as irregular power generation from renewable sources or unexpected fluctuations in energy production. To bridge the gap between supply and demand, electricity is procured from the grid. This allows for the continuous fulfillment of consumers' electricity needs, even during periods when RES are

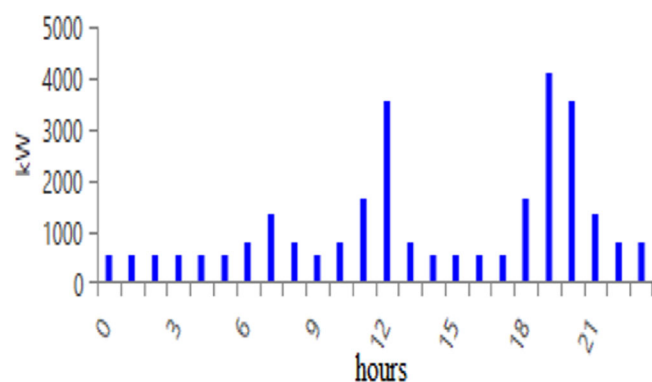


FIGURE 6 Daily load profile forecast for 2030.

TABLE 3 Specification of system components.

Component	Size	Capital cost (\$)	Replacement cost (\$)	O&M cost(\$)	Life time (year)
Solar photovoltaic	1 kW	883	529.8	8.83	25
Wind Turbine	3 kW	4065	2844.5	81.3	20
Battery	3.123 kWh	800	560	8.0	4
Converter	1 kW	800	560	8.0	15

unable to generate the desired amount. By integrating the grid as a reliable backup, the study aims to provide uninterrupted power supply and improve the overall stability and reliability of the electricity system. Table 4 provides a summary of the technical details and assumed parameters used in the study.

2.5 | Sensitivity analysis

In the sensitivity analysis, multiple optimizations are performed with a variety of inputs to evaluate the impact of variables over which the designer has no control, such as solar radiation, and wind speed data.^{59,60} Furthermore, sensitivity analysis aids in understanding how the optimization results change as input variables change.

2.5.1 | Techno economic analysis

A techno-economic analysis is a tool for determining the economic effectiveness of a process, product, or service.^{61,62} In general, the profitability of a project is determined by analyzing economic indicators such as NPC, COE, and Payback Period (PB). These economic indicators, which are crucial decision-making tools,⁶³ for investors, governments, and donors, were analyzed in the current study.

2.5.2 | Total net present cost (TNPC)

The TNPC of a system is the present value of all costs associated over the system's lifespan minus the present value of all income earned over the system's lifetime.⁶⁴ Costs include capital costs, replacement costs, O & M costs, fuel costs, emissions penalties, and the costs of buying power from the grid. Salvage value and grid sales revenue are included. HOMER calculates the total NPC by summing the total discounted cash flows in each year of the project lifetime using the below equation.

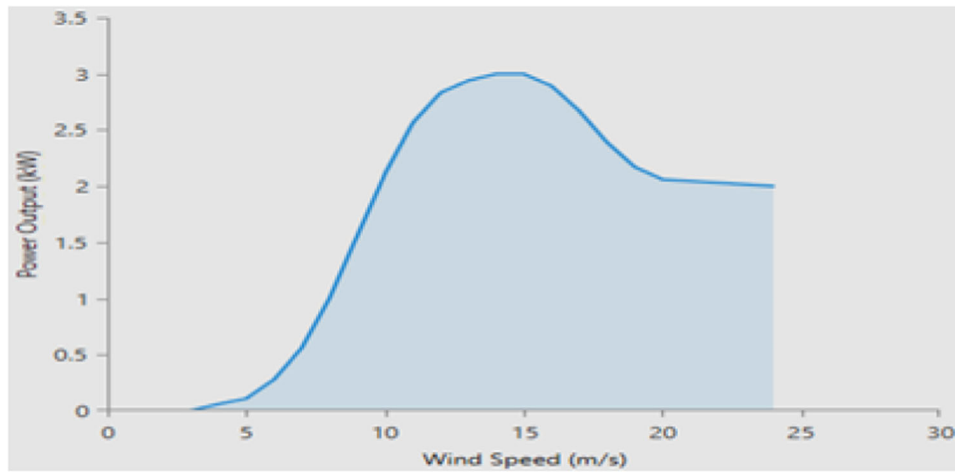


FIGURE 7 Power versus wind speed distribution curve.

TABLE 4 Different technical details and assumptions parameters used in the study.

Parameters	Value
System lifetime (year)	25
Annual discount rate (%)	5
Homer pro software version	3.14.2
Minimum renewable fraction (%)	40
Maximum annual capacity shortage (%)	10
Simulation time step (minutes)	60
Load factor	0.17
Random variability of load day to day (%)	10
Random variability of load time step to time step (%)	20
Currency (US D)	\$
Cost of electricity (\$)	0.114
Cost of returning power to the grid (\$)	0.05
Operating system type (bits)	64
Processor (GHz)	1.99
RAM (GB)	8.0
hp, intel Laptop (piece)	1
Core	i7
Generation	8

$$\begin{aligned}
 NPC = CF_0 + \left\{ \frac{CF_1}{(1+i)^1} + \frac{CF_2}{(1+i)^2} + \frac{CF_3}{(1+i)^3} \right. \\
 \left. + \dots + \frac{CF_N}{(1+i)^N} \right\} NPC = CF_0 \\
 + \sum_{i=1}^N \frac{CF_i}{(1+i)^i},
 \end{aligned} \quad (1)$$

where CF_0 is initial capital cost (USD). CF_t is the cash flow of t -year (USD). t is the number of years (year). i is the annual real interest rate (%). N is project duration (years).

2.5.3 | COE

Homer software defines the COE as the system's average cost per kWh of useful electrical energy generated.⁶⁵ It is determined by dividing the total annualized cost (TAC) by the total annualized useful electrical energy generated, as illustrated in the below equation

$$\begin{aligned}
 COE &= \frac{TAC}{E_{Served}} \\
 &= \frac{\text{Total anualised cost} \left(\frac{\$}{\text{year}} \right)}{\text{Annual load served (kWh/year)}},
 \end{aligned} \quad (2)$$

$$\text{Total anualised cost (TAC)} = NPC \times CRF, \quad (3)$$

$$\begin{aligned}
 \text{Annual load served}(E_{served}) &= E_{primary} + D_{efferer} \\
 &+ E_{grid,sales},
 \end{aligned} \quad (4)$$

where TAC is the annualized value of the NPC in \$/year. E_{served} is the total electrical load served (primary load, deferred load, and energy sales to the grid) in kWh/year. CRF is the capital recovery factor. i is the real interest rate (%) N is the project lifetime (years).

2.5.4 | PB

The duration required to recoup the investment made in the installation of a plant is termed as the PB.⁶⁶ The

payback is an indication of how long it would take to repay the difference in investment costs between the optimized and basic case systems^{67,68} as presented in the below equation.

$$\text{PBP} = \frac{\text{Initial investment}}{\text{Cash flow per period}}. \quad (5)$$

3 | RESULTS AND DISCUSSION

This section presents the obtained results and provides a comprehensive analysis of the optimal design and analysis of a grid-connected hybrid renewable energy system (HRES) using HOMER Pro as a simulation tool. The sub-section in this section offers an in-depth analysis of the results and stimulates discussion on the grid-connected HRES design for Tumbatu Island, contributing to the progression of sustainable energy solutions in similar island communities.

3.1 | Simulation and optimization of various hybrid energy systems

A total of 7208 solutions were simulated by Homer Pro, of which 1536 were feasible. Homer then optimized the possible options and ranked them into the top six solutions, which are shown in Table 5a. The results obtained from Table 5a indicate that Configuration A is the most favorable choice among the analyzed hybrid renewable energy setups. Configuration A, which includes PV, wind, and grid components, stands out due to its highly competitive NPC of \$4,003,851 making it a cost-effective solution. Furthermore, Configuration A

boasts a short PB of 3.79 years, indicating a quick return on the initial investment. Although its COE is slightly higher at \$0.09 per kWh compared to Configuration B, the overall balance of cost-effectiveness and a swift payback establishes Configuration A as the optimal choice. The inclusion of both PV and wind sources, along with grid connectivity, ensures a reliable and consistent energy supply. Decision-makers can utilize these findings to prioritize Configuration A when seeking an economically efficient and promptly rewarding hybrid renewable energy solution. While Configuration A emerges as the best choice among the analyzed hybrid renewable energy setups, Configuration B, consisting solely of PV panels with grid connectivity, also presents notable attributes. With a NPC of \$4,005,868 and a slightly longer PB of 4.88 years, Configuration B represents a competitive option. The absence of wind or battery components simplifies the system design, which may be appealing in scenarios where simplicity and reduced upfront investment are prioritized.

3.2 | Analysis of voltage profile

The stability of the electric grid plays a crucial role in effectively incorporating renewable energy systems. This section focuses on verifying and analyzing the voltage profile in a grid-connected HRES. Three different scenarios (Case 1, Case 2, and Case 3) are investigated, with each case representing different ranges of inverter output power. Validation of the results was conducted using three different cases. In Case 1, the inverter output power ranged from 376.0 to 364.7 kW. In Case 2, the power varied from 200 to 194 kW. Case 3 had a power range of 193 to 188 kW, as indicated in Table 5b. The inverter capacity of 376 kW was presented in Table 5a, and the number of inverters per phase was calculated

TABLE 5a Optimization results.

	Sizes of various components				Grid	Net present cost (\$)	Cost of energy (\$)	Payback period (year)
	PV (kW)	Wind turbines (quantity)	Converter (kW)	Battery (kW)				
A Photovoltaic (PV)/wind/grid	685	10	376	–	ON	4,003,851	0.09	3.79
B PV/grid	750	–	424	–	ON	4,005,868	0.088	4.88
C PV/wind/battery/grid	731	5	392	1	ON	4,007,771	0.089	4.52
D PV/battery/grid	743	–	409	17	ON	4,028,774	0.089	4.2
E Wind/grid	–	200	–	–	ON	4,491,925	0.091	n/a
F Wind/battery/grid	–	200	12	17	ON	4,529,434	0.092	n/a

TABLE 5b Simulation results.

Month	Energy purchased (kWh)	Energy sold (kWh)	Net energy purchased (kWh)	Inverter output power per day (kW)		
				Case 1	Case 2	Case 3
January	126,540	34,931	91,609	376.03	200.00	194.00
February	113,774	33,355	80,419	368.12	195.53	191.42
March	135,314	30,776	104,539	373.44	198.88	192.80
April	107,119	29,114	78,005	373.00	197.32	192.20
May	123,863	25,088	98,775	372.00	196.74	191.77
June	123,910	24,020	99,890	364.71	195.00	188.00
July	124,540	27,049	97,491	367.23	196.40	189.50
August	136,937	27,570	109,367	372.83	196.21	190.63
September	130,550	31,211	99,340	373.00	196.00	190.33
October	122,927	34,184	88,743	375.58	198.90	190.00
November	131,118	30,005	101,113	372.37	195.30	191.32
December	141,093	29,861	111,233	374.80	198.00	193.88
Annual	1,517,686	357,163	1,160,522	4463.11	2364.28	2295.85

using the below equation. Detailed specifications of the inverters can be found in Appendix A.

$$\begin{aligned} \text{Inverters per phase} &= \frac{\text{Inverter capacity}}{3 \times \text{Inverter rating}} \\ &= \frac{376}{3 \times 25} = 5, \end{aligned} \quad (6)$$

$$\begin{aligned} \text{Current per phase } (I_p) \\ &= \text{Number of Inverter per phase} \\ &\quad \times \text{Maximum current.} \end{aligned} \quad (7)$$

Total power in three phase balanced load is demonstrated in the below equation.⁶⁹

$$P = 3I_p V_p \cos \theta, \quad (8)$$

where P is the power per phase, I_p is the phase current, V_p phase voltage and $\cos \theta$ is the power factor = 0.8. Equation (8) was employed to derive the output voltage profile of RES preintegration into the grid, as presented in Tables 6–8. Upon integration of PV and wind on the LV side of the transformer, the transformer characteristic transitions from step-down to step-up. The elevated voltage on the HV side is computed using the transformer ratio in the below equation, yielding results shown in Tables 6–8.

$$\frac{N_H}{N_L} = \frac{V_H}{V_L} = k, \quad (9)$$

where N_H and N_L are the number of turns of HV and LV side respectively.

V_H and V_L are the phase voltage at HV and LV sides, respectively obtained in Table 1. The value of k can be computed using voltages in Table 1. The standard voltage in Tanzania is 230 volts, alternating at 50 cycles (Hertz) per second,⁶⁹ this is the same as or similar to most countries throughout the world. The HV is calculated using Equation (10), and the results are displayed in Table 6. The inverter ensures that the grid receives the standard voltage, even in cases where the voltage is higher. The voltage from the solar PV and wind turbine will be synchronized at a common bus bar (230 V). Due to the high potential of the voltage generated by solar PV, it will be fed back into the grid, as depicted in Tables 6–8.

$$V_E = V_I k, \quad (10)$$

where V_I is the inverter output voltage and V_E is the elevated voltage on the HV side of the transformer.

Table 6 provides a comprehensive summary of the voltage profile of the suggested renewable energy system on Tumbatu Island for Case 1, while Figures 8 and 9 depict the voltage profile of renewable energy both before and after integration with the grid on the LV side, respectively.

TABLE 6 Voltage profile of proposed system in Tumbatu Island (Case 1).

Month	Renewable energy (kV)	Tumbatu feeding voltage (kV)	Remaining voltage after feeding (kV)	Voltage from the Inverter to the low voltage side of the grid (kV)	Stepped up the voltage to the high voltage side (kV)
January	0.870	0.23	0.655	0.23	31.19
February	0.852	0.23	0.637	0.23	31.19
March	0.864	0.23	0.646	0.23	31.36
April	0.863	0.23	0.645	0.23	31.36
May	0.861	0.23	0.643	0.23	31.36
June	0.844	0.23	0.626	0.23	31.36
July	0.850	0.23	0.632	0.23	31.36
August	0.863	0.23	0.645	0.23	31.36
September	0.863	0.23	0.645	0.23	31.36
October	0.869	0.23	0.651	0.23	31.36
November	0.862	0.23	0.647	0.23	31.23
December	0.868	0.23	0.653	0.23	31.23

TABLE 7 Voltage profile of proposed system in Tumbatu Island (Case 2).

Month	Renewable energy (kV)	Tumbatu feeding voltage (kV)	Remaining voltage after feeding (kV)	Voltage from the Inverter to the low voltage side of the Grid (kV)	Stepped up the voltage to the high voltage side (kV)
January	0.463	0.23	0.233	0.230	31.19
February	0.453	0.23	0.223	0.223	30.24
March	0.460	0.23	0.230	0.230	31.36
April	0.457	0.23	0.227	0.227	30.95
May	0.455	0.23	0.225	0.225	30.68
June	0.451	0.23	0.221	0.221	30.14
July	0.454	0.23	0.224	0.224	30.54
August	0.454	0.23	0.224	0.224	30.54
September	0.454	0.23	0.224	0.224	30.54
October	0.460	0.23	0.230	0.230	31.36
November	0.452	0.23	0.222	0.222	29.87
December	0.458	0.23	0.228	0.228	29.87

Table 6 reveals the variations in combined output voltage from solar and wind observed across different months, ranging from 0.844 kV in June to 0.870 kV in January. Additionally, the feeding voltage on Tumbatu Island remains stable at 0.23 kV. Fluctuations in the remaining voltage after feeding, ranging from 0.626 to 0.655 kV,

indicate dynamic changes occurring within the local grid. The voltage from the inverter to the LV side remains constant at 0.23 kV, ensuring a reliable conversion process. Furthermore, the stepped-up voltage to the high-voltage side improved to 31.23 kV at peak load, indicating efficient long-distance power transmission. It can be concluded that

TABLE 8 Voltage profile of proposed system in Tumbatu Island (Case 3).

Month	Renewable energy (kV)	Tumbatu feeding voltage (kV)	Remaining voltage after feeding (kV)	Voltage from the Inverter to the low voltage side of the grid (kV)	Stepped up the voltage to the high voltage side (kV)
January	0.449	0.23	0.219	0.219	29.70
February	0.443	0.23	0.213	0.213	28.89
March	0.446	0.23	0.216	0.216	29.45
April	0.445	0.23	0.215	0.215	29.32
May	0.444	0.23	0.214	0.214	29.18
June	0.435	0.23	0.205	0.205	27.95
July	0.439	0.23	0.209	0.209	28.50
August	0.441	0.23	0.211	0.211	28.77
September	0.441	0.23	0.210	0.210	28.64
October	0.440	0.23	0.210	0.210	28.64
November	0.443	0.23	0.213	0.213	29.92
December	0.449	0.23	0.219	0.219	29.74

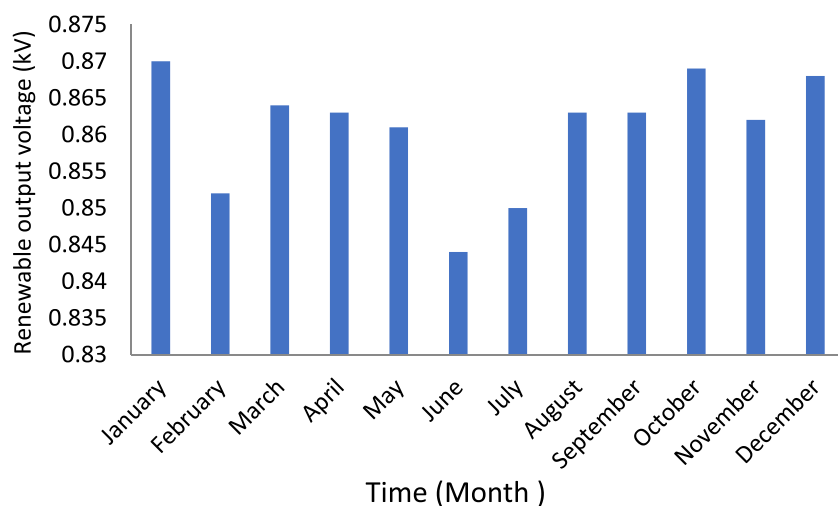


FIGURE 8 Voltage profile of renewable energy before integrated to the grid on the LV side.

the higher the renewable generation the higher the power output of hybrid system and the greater the improvement in the voltage profile of the grid. This finding is concurred with Ali et al.²² who stated that the higher the solar radiation, the higher the power output of the solar generator and the greater the improvement in the voltage profile of the electrical system. These findings offer valuable insights into energy generation patterns, system stability, and the potential for sustainable electricity supply⁷⁰ on Tumbatu Island.

In Table 7, the voltage profile of the proposed renewable energy system on Tumbatu Island (Case 2) is presented for a period of 12 months. Solar and wind

generation exhibit slight fluctuations, ranging from 0.450 to 0.463 kV. The feeding voltage remains constant at 0.23 kV, showing consistent power input. The overall voltage level after feeding experiences minimal impact, ranging from 0.220 to 0.233 kV, indicating a stable grid. The transmission voltage to the LV side remains consistent at 0.220 to 0.230 kV, ensuring reliable power flow. The stepping up of voltage to the high-voltage side is efficient, ranging from 29.87 to 31.36 kV. In Case 2, the system demonstrates stable and dependable performance, which is vital for maintaining a consistent power supply to Tumbatu Island. In Table 8, a comprehensive monthly breakdown of the voltage profile for the

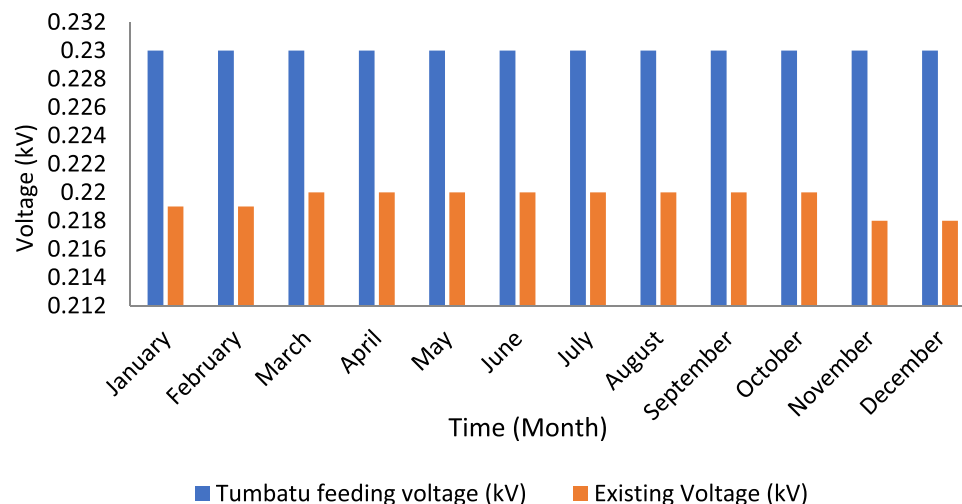


FIGURE 9 Voltage profile of renewable energy when integrated into the Grid on the low voltage side.

proposed renewable energy system on Tumbatu Island (Case 3) is presented. Notable aspects include a consistent feeding voltage of 0.23 kV, indicating stability at the point of renewable energy entry. The remaining voltage after feeding provides valuable insights into the available grid capacity, while the voltage from the inverter to the LV side and the stepped-up voltage to the HV side highlight the operational efficiency of the system. Monthly variations in the combined output voltage from solar and wind sources are also clearly visible. These findings are essential for assessing system stability, efficiency, and guiding decision-making processes to ensure a sustainable and reliable energy supply on Tumbatu Island as concluded by Dudi and Sharma.⁷¹

Among the three cases, Case 2 emerges as the most stable and reliable option. It exhibits minimal fluctuations in output voltage, maintains a steady feeding voltage, and consistently delivers dependable power to both the LV and HV sides. The efficient stepping up of voltage enhances the effectiveness of the long-distance power transmission system. Overall, the system in Case 2 appears to be well-suited for ensuring a consistent and reliable power supply to Tumbatu Island, making it the preferred option. However, despite Case 3 demonstrating operational efficiency in terms of maintaining voltage stability and increasing voltage, the monthly fluctuations observed in the combined output voltage could potentially affect the overall efficiency of the system. In situations where predictability is of utmost importance, as is the case of Tumbatu Island, this variability may present difficulties in effectively managing the power supply. Similarly, despite the variations in combined output voltage in Case 1 falling within an acceptable range, the monthly fluctuations (ranging from 0.844 to 0.870 kV) have the potential to introduce uncertainties in

energy generation patterns. This variability may impact the consistency of the power supply and potentially pose challenges in meeting demand during specific months.⁷² It is clear from the results that ensuring system stability and efficiency is key in the creation of sustainable renewable energy solutions^{73,74} for remote areas like Tumbatu Island.

3.3 | Voltage profile improvement results

To improve the voltage profile of the transmission line (the grid), the/Wind/Battery/Converter system and the utility grid were designed using the HOMER Pro software. Through simulation, it was determined that the optimal solution involved the combination of PV/Wind and the grid. This solution resulted in an improved voltage profile, as evidenced by the data presented in Tables 5a and 5b. The results of the simulation and optimization for various configurations of a renewable energy system, consisting of PV panels, wind turbines (WT), converters, batteries, and grid connection, are summarized in Table 5a. The results in Table 5a for different renewable energy configurations are summarized, presenting information on component sizes, NPC, COE, and PBP. It is important to highlight that configuration B (PV/GRID) stands out as an economical option, with the lowest NPC, favorable COE, and a reasonable PBP of 4.88 years. This particular configuration, which relies on PV panels and grid connectivity, emphasizes a well-balanced and economically viable approach to maintaining a consistent electricity supply.³⁶

Table 5b presents the simulation results for three different cases (Case 1, Case 2, and Case 3) across various

months. In all cases and months, the net energy purchased is positive, indicating a consistent trend of buying more energy than selling. This suggests a reliance on external sources to meet energy demands, which for this study is the grid. The inverter output power per day for each case varies across the months. Case 1 consistently has the highest output, followed by Case 2 and Case 3. This suggests that the system in Case 1 is designed to generate more power, while Case 3 has the lowest output. In terms of electricity availability, the positive values in the net energy purchased column signify a dependence on the grid, to fulfill energy requirements. This implies that the PV/Wind system is not entirely self-sufficient and relies on purchasing additional energy from the grid to meet the demand. The differences in inverter output power per day between the cases indicate varying levels of energy generation capacity, with Case 1 being the most productive. The optimization results in Table 5b indicate a consistent pattern of relying on purchased energy from the grid, with variations in inverter output power per day across different cases. The results exhibit higher performance of RES in stabilizing the grid when there is insufficiency in terms of electricity supplied by the grid. In comparing with previous studies, Ehteshami and Chan⁷⁵ conducted a comprehensive study on renewable energy systems and concluded that ensuring system stability and improving efficiency are vital for the long-term sustainability of renewable energy integration. In a research paper by Hoang and Nguyen,⁷⁶ they examined the relationship between system stability, efficiency, and sustainable renewable energy. Their findings align with the importance of maintaining stable and efficient systems for achieving sustainable energy goals. When comparing the three scenarios for grid integration, it can be concluded that Case 2 is the most preferable. It achieves a good

balance by providing a reasonable output power from the inverter each day while reducing the amount of energy purchased from external sources. This indicates a higher capacity for generating energy while minimizing dependency on outside suppliers. Therefore, it can be inferred that Case 2 is a potentially more sustainable option for grid integration when compared to the other cases.

Furthermore, the integration of solar PV and wind turbines led to an increase in harvested voltage, a reduced reliance on the grid, and an enhanced capacity for integration into the grid, as indicated in Table 6. Before the integration of RES, there was a noticeable decline in voltage levels, as illustrated in Table 1, Figures 10 and 11. This decline was observed in both the LV and HV sides of the transformer, indicating an overall improvement in voltage levels. These findings remained consistent across different inverter outputs, ranging from 376.03 to 364.7 kW (Case 1), 200 to 195 kW (Case 2), and specifically in the months of December and January, 194 to 188 kW in Case 3, as depicted in Figure 12. When compared the finding with existing literatures, some studies, such as Elshahed,⁷⁷ achieved an improvement in voltage profile by 6.3%, while²⁴ improved voltage profile by 1.97%. Additionally,⁷⁸ enhanced the voltage profile by 8%. In the present study, the voltage profile improved from 29.6 to 31.23 kV (05%). The difference in findings depends on the size of the load, availability of resources in different climatic conditions and areas.⁷⁹

3.4 | Economic details of the optimized system

Table 5b illustrates that for the selected system A, the annual energy purchased from the grid amounts to

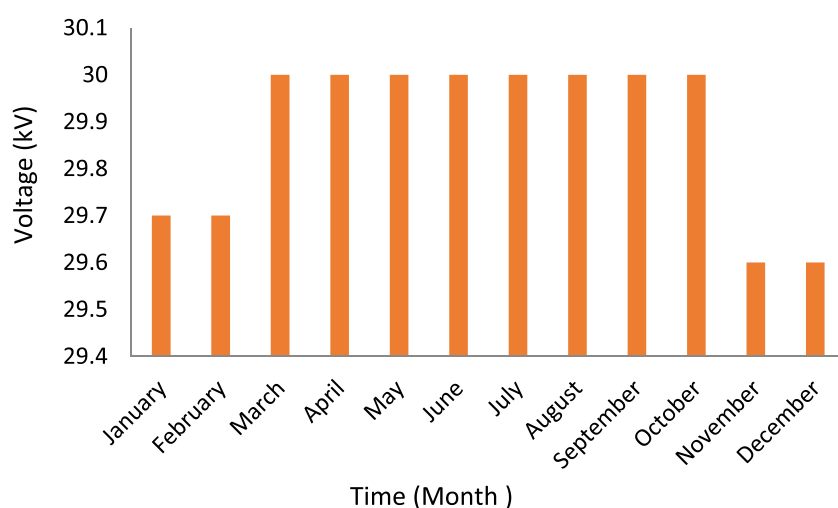


FIGURE 10 Existing voltage profile for Tumbatu Island on the high voltage side.

FIGURE 11 Existing voltage profile for Tumbatu Island on the low voltage side.

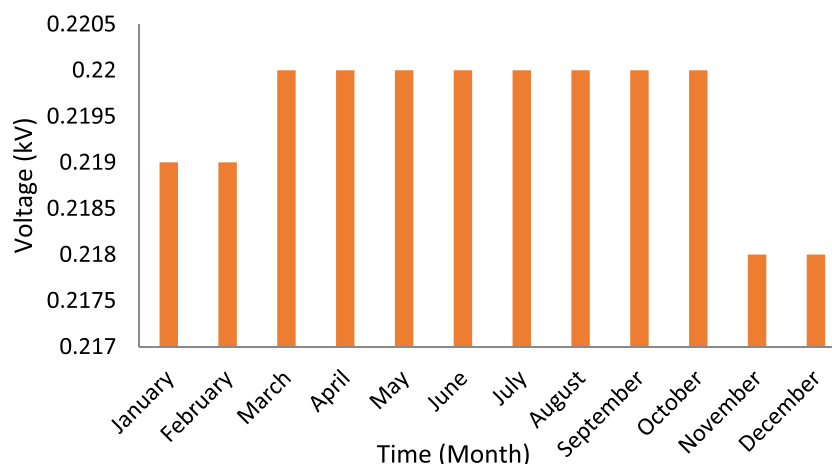
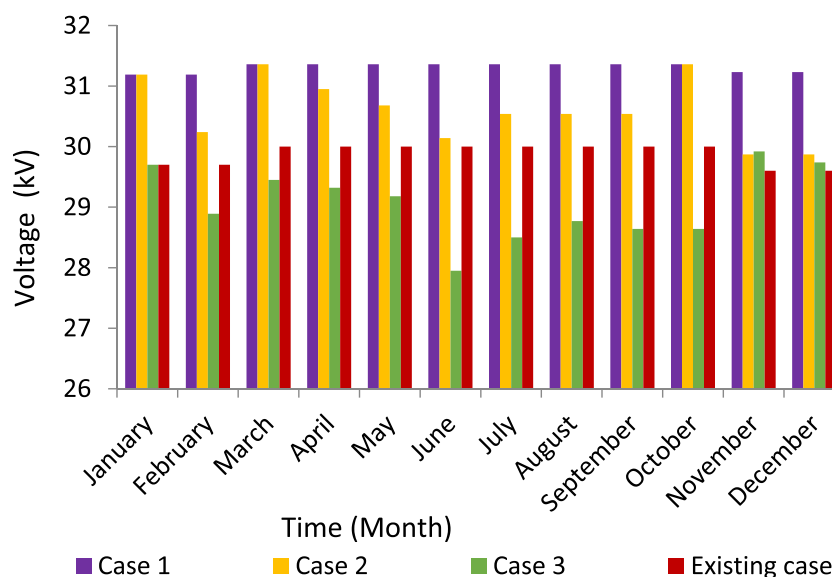


FIGURE 12 Voltage profile of existing and proposed on the high voltage side.



1517.686 MWh, while the annual energy sold back to the grid is 357.163 MWh. Additionally, the minimum electrical energy sold to the grid in June is 24.020 MWh, whereas the maximum electrical energy sold to the grid in January is 34.931 MWh. These results indicate that renewable energy, especially solar PV, produced the least amount of electricity in June and the highest amount in January. Variations in electricity production by solar PV and the wind turbine are depicted in Figures 13 and 14 respectively, while Figure 15 displays the variation in output power via the system converter throughout the year. Figure 16 illustrates the fluctuations in primary load, electricity sales to the grid, and electricity purchases from the utility grid for selected days in January, providing a clear visual representation of the energy transactions. The red lines represent the primary load, black lines depict electricity purchases from the utility grid, and blue lines show the electricity sold back to the grid. As the proposed HRES does not incorporate storage

technologies, excess electricity generated by solar PV and the wind turbine can be efficiently sold back to the utility grid. The chosen hybrid renewable energy configuration A, encompasses components such as solar PV, wind turbine, grid, and converter, all contributing to its economic viability. Each component in the system incurs costs related to capital, operational, maintenance, and replacement expenses. The electricity production of the HRES stems from these diverse components. To visually illustrate the cost parameters of the components within optimized system A, Figure 17 provides a clear representation.

3.5 | Sensitivity analysis

In this study, a comprehensive sensitivity analysis was conducted to investigate the impact of several sensitive variables on the cost parameters of the HRES. Table 9

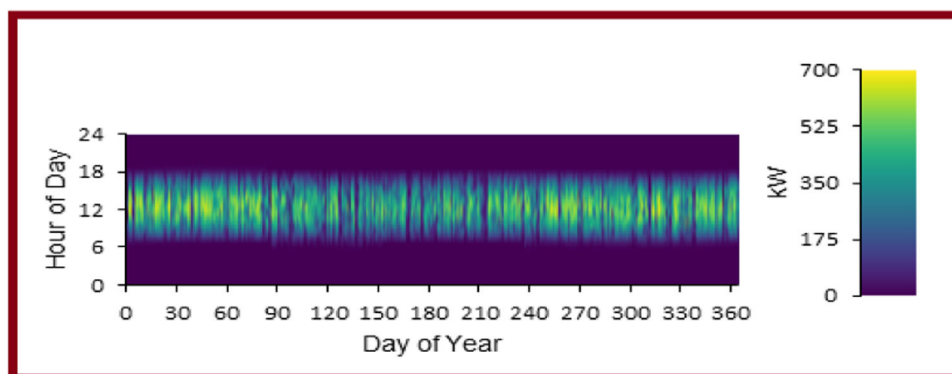


FIGURE 13 Solar photovoltaic output power production throughout the year.

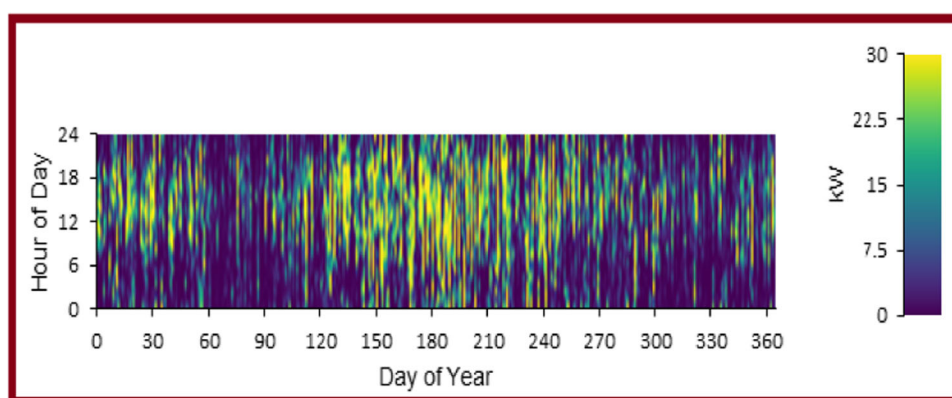


FIGURE 14 Wind Turbine output power production throughout the year.

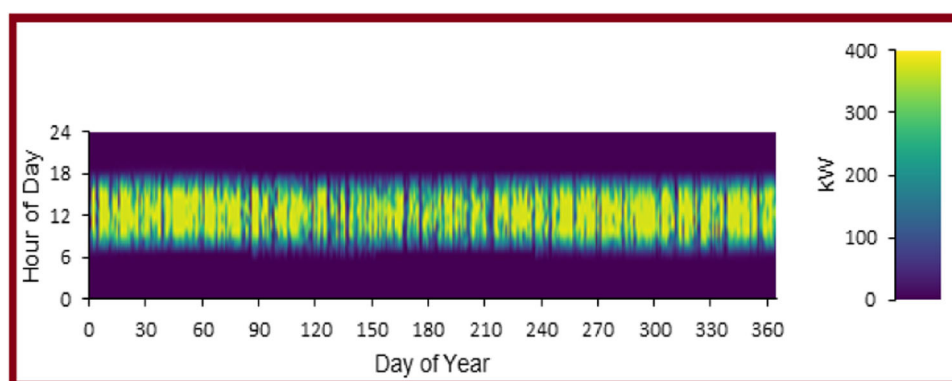


FIGURE 15 System converter variation of output power throughout the year.

presents the different sensitive parameters and their corresponding values that were evaluated in the analysis. The sensitivity parameters included wind speed, with values of 6.24, 6.5, and 7 m/s, and solar radiation, with values of 5.57, 5.8, 6, and 6.6 kWh/m²/day. Additionally, the scaled annual average electric load was considered, with values of 6000, 9000, 10,000, and 11,000 kWh, as well as the nominal discount rate, which was assessed at

5% and 4%. By varying these sensitive variables, the study aimed to comprehensively analyze how changes in wind speed, solar radiation, electric load, and discount rate could influence the cost parameters of the HRES. This analysis is important for understanding the system's performance and financial viability under different conditions and uncertainties, providing valuable insights for optimal system design and decision-making.

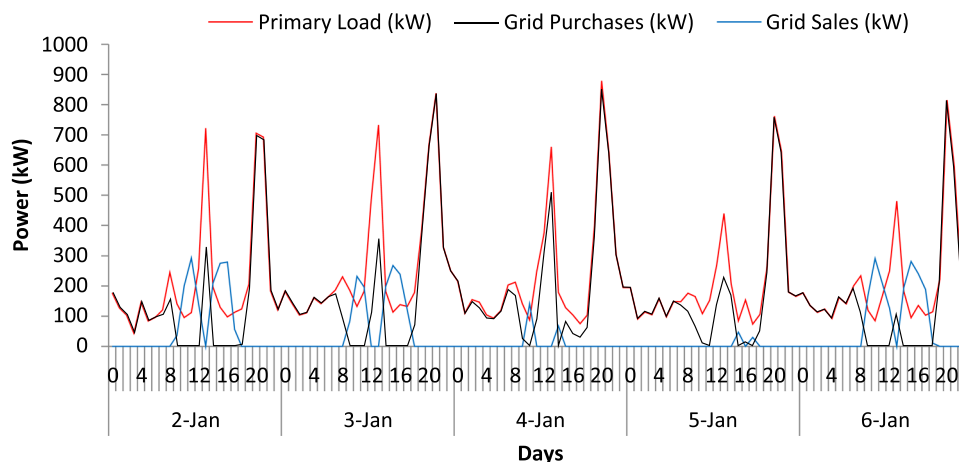


FIGURE 16 Variations of power output from scenario A with grid purchase and sales.

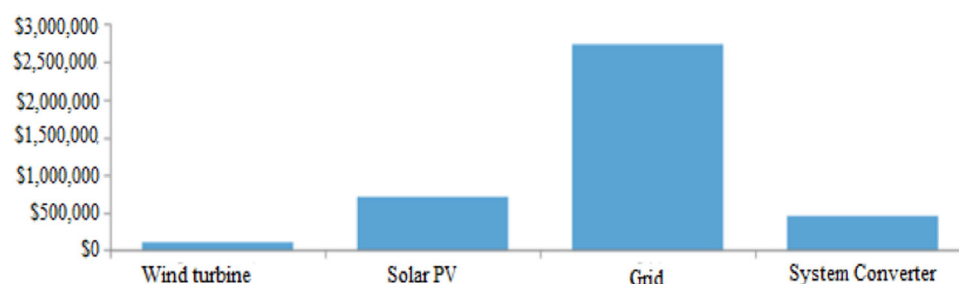


FIGURE 17 Summary of system cost by component.

TABLE 9 Various sensitive variables with various values.

Parameter	Wind	Solar	Load	Nominal discount rate
Sensitivity parameter	Speed	Radiation	Scaled annual average electric load	Percentage of discount rate
Values	6.24, 6.5, 7	5.57, 5.8, 6, 6.6	6000, 9000, 10,000, 11,000	5, 4

3.5.1 | Sensitivity analysis of solar radiation on cost parameters

The variation in solar radiation levels plays a significant role in determining the electrical energy output from the solar PV system. To investigate this impact, three distinct solar radiation sensitivity values were input into the Homer Pro software, allowing for a comprehensive assessment of how solar radiation influences the cost parameters. The information regarding radiation, COE, and TNPC is presented in a graphical format in Figure 18. As radiation levels increase from 5.57 to 6.6 kWh/m²/day, a noticeable pattern emerges in both the COE and TNPC columns. The COE decreases from 0.0897 to 0.0785 \$/kWh, indicating a reduction in the COE. Simultaneously, the TNPC decreases from \$4,000,000 to \$3,880,000 suggesting a decrease in the TNPC. In terms of electricity availability, interpreting results in Figure 18 reveals that higher

radiation levels contribute to more favorable economic outcomes. The decreasing trend in both COE and TNPC implies that as radiation increases, the cost of generating electricity decreases, leading to an improvement in overall economic feasibility.^{80,81} This could be attributed to increased efficiency or higher energy output associated with greater radiation levels, resulting in electricity being more readily available at a lower cost. Figure 18 provides a visual representation of this relationship, highlighting how changes in solar radiation levels can influence the overall cost parameters of the HRES.

In previous studies, several researchers have concurred with the findings of this study based on solar radiation on cost parameters (COE and TNPC). For example,⁸² reported similar cost reductions in their research, stating, “As radiation levels increase, a noticeable decrease in the COE is observed.” In line with this,⁸⁰ conducted a study where they observed a decrease in the

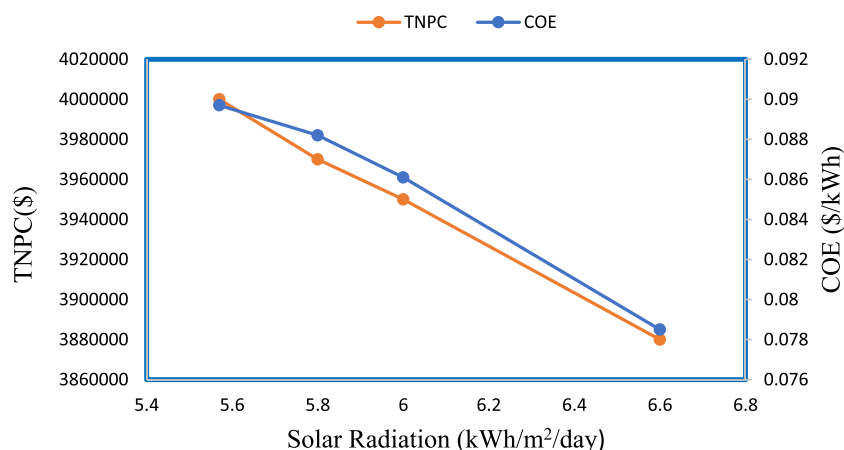


FIGURE 18 Impact of solar radiation on total net present cost (TNPC) and cost of energy (COE).

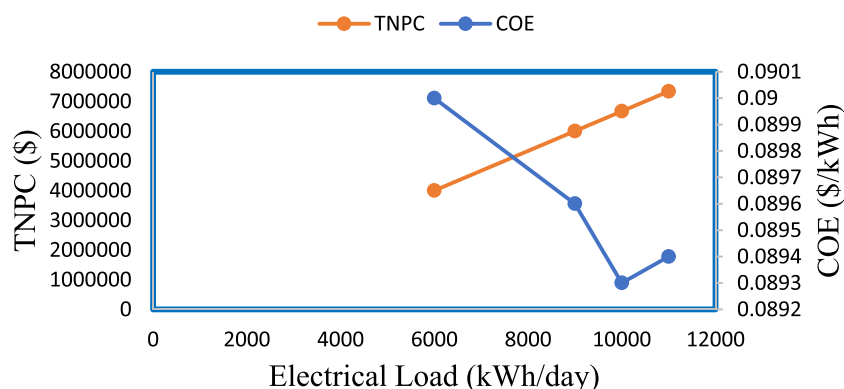


FIGURE 19 Variation of electric load on cost parameter. COE, cost of energy; TNPC, total net present cost.

overall cost of the HRES with higher radiation levels, affirming that “increased radiation levels contribute to more favorable economic outcomes.” In a separate study,³⁶ emphasized the positive relationship between radiation levels and economic feasibility, stating that “higher radiation levels lead to improved economic viability.” Similarly, Haidar et al.,⁸³ found that as radiation increases, there is a decrease in the cost of generating electricity. They stated, “The trend indicates that as radiation levels rise, the COE decreases.” Lastly, Buni et al.⁸⁴ highlighted the positive impact of higher radiation levels on energy production, stating that “increased radiation levels result in higher energy output and improved efficiency.” These various studies align with the findings presented in this study, further supporting the significance of solar radiation levels in determining the electrical energy output and cost parameters of solar PV systems.

3.5.2 | Sensitivity analysis of electric load on cost parameters

The sensitivity analysis conducted in this study also examined the impact of electric load variations on the

optimized hybrid system's TNPC and COE. The investigation revealed interesting findings regarding the behavior of these cost parameters in response to changes in electrical load. As illustrated in Figure 19, it was observed that as the electrical load increases, the NPC values also increase. However, the COE initially decreases from 6000 to 10,000 kWh/day and then starts increasing again from 10,000 to 11,000 kWh/day. Furthermore, the sensitivity analysis allowed us to examine the effect of electric load changes on the individual components of the hybrid system. It was noted that while the quantity of wind turbines remained constant at 10 units in the original design, the capacity of solar panels increased in correlation with the increase in electric load, as depicted in Figure 20. This result indicates that the solar panel capacity is directly affected by variations in the electrical load in this particular scenario. The findings from this sensitivity analysis shed light on the dynamic relationship between electric load, cost parameters, and the sizing of renewable energy components in the optimized hybrid system. This finding aligns with the findings of a study conducted by Malik et al.,⁸¹ who investigated the decrease of COE and rise of TNPC as electric load increases. Similarly, Sharma et al.³⁶ observed when electrical load increases,

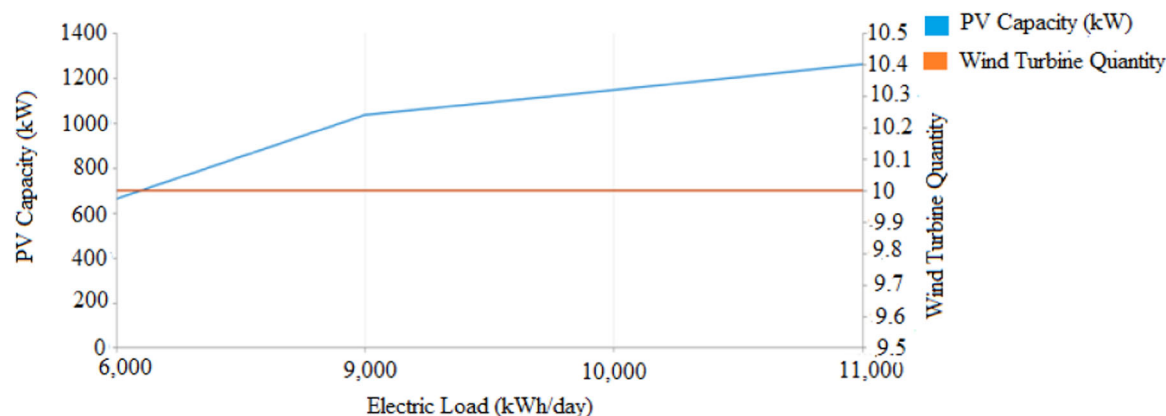
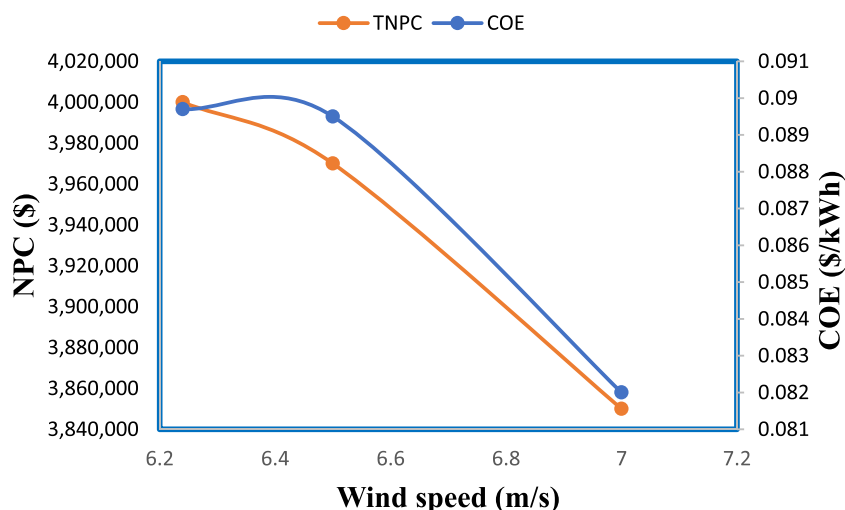


FIGURE 20 Variation of electric load on photovoltaic (PV) cost and wind turbine quantity.

FIGURE 21 Impact of wind speed on cost parameter. COE, cost of energy; NPC, net present cost.



COE and TNPC increase. Furthermore, Manyama⁵¹ observed the consistence in wind turbine quantity as the electric load rises.

3.5.3 | Sensitivity analysis of wind speed on cost parameter

Wind speed is the most important aspect of producing electricity using wind turbines. In this case, the wind speeds tested were 6.24, 6.5, and 7 m/s. The result revealed that both TNPC and COE decrease as wind speed increases as presented in Figure 21. This finding is consistent with the findings of a study conducted by Malik et al.⁸¹ Stated if the wind speed is increased COE and TNPC decline. In a separate study,⁸⁵ emphasized the positive relationship between wind speed levels and economic feasibility, stating that increasing average wind speed levels lead to decrease in system costs.

3.5.4 | Sensitivity analysis of nominal discount rate on cost parameters

In Tanzania, the nominal discount rate holds significant importance in economic evaluations and investment decisions. To comprehensively assess its impact on the cost parameters of optimized HRES, a sensitivity analysis is conducted using two different discount rates: 5% and 4%. The findings, depicted in Figure 22, provide valuable insights into how variations in the nominal discount rate influence the cost parameters. As the discount rate decreases from 5% to 4%, the NPC experiences a corresponding increase, while it decreases as the discount rate rises. This phenomenon can be attributed to the time value of money; lower discount rates place greater emphasis on future cash flows, thus leading to higher present costs. On the other hand, higher discount rates place more weight on present cash flows, resulting in lower present costs.

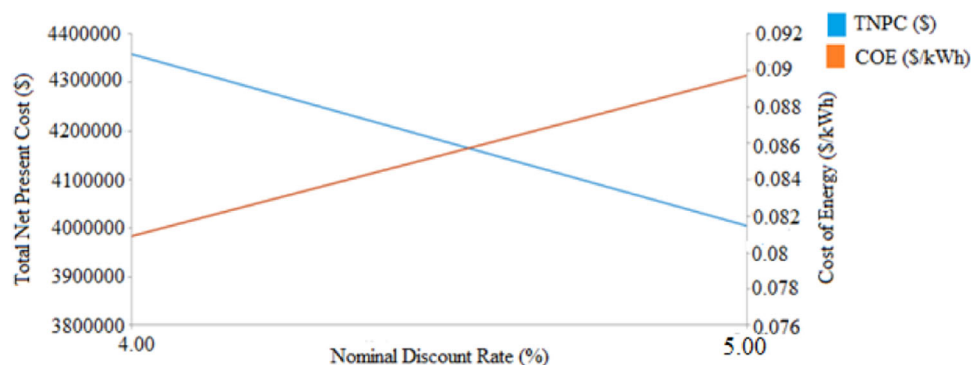


FIGURE 22 Impact of nominal discount rate on cost parameters. COE, cost of energy; TNPC, total net present cost.

Sensitivity	Architecture								Cost			
NominalDiscountRate (%)									NPC (\$)	COE (\$)	Operating cost (\$/yr)	Initial c (\$)
4.00									\$4.36M	\$0.0809	\$161,415	\$1.19M
5.00									\$4.00M	\$0.0897	\$174,437	\$946,3

FIGURE 23 Sensitivity analysis of nominal discount rate.

The COE, on the other hand, exhibits an inverse relationship with the nominal discount rate. As the discount rate decreases, the COE decreases as well, and vice versa. This outcome is a direct consequence of the interplay between the discount rate and the Levelized COE. A lower discount rate reduces the cost of financing, leading to lower COE, while a higher discount rate has the opposite effect, raising the COE. Based on these findings, it is evident that the choice of discount rate has a profound impact on the economic evaluation of renewable energy projects. The selection of an appropriate discount rate is crucial for accurately assessing the viability and attractiveness of investment opportunities. A lower discount rate may be favored when evaluating long-term and sustainable energy projects, as it values future benefits more highly and encourages the development of renewable energy infrastructure.

The optimization results, as presented in Figure 23, provide a comprehensive overview of the sensitivity analysis with the 4% discount rate. It is essential to consider such variations in the discount rate during project planning and decision-making processes to ensure sound economic evaluations and foster the sustainable growth of renewable energy initiatives in Tanzania. The nominal discount rate significantly impacts the cost parameters of HRESs. Understanding the dynamics between the discount rate, NPC, and COE is vital for making informed investment decisions and maximizing the economic benefits of renewable energy

projects. With a clear understanding of these relationships, stakeholders can chart a path toward a greener, more sustainable, and economically viable future for Tanzania's energy landscape. The findings of nominal discount rate on cost parameters (COE and TNPC) for this study concurred with different researchers. For instance,⁸⁶ revealed that when nominal discount rate increases, the COE increases while TNPC decreases. Also, this finding of nominal discount rate lies with the study,⁸⁷ that COE is proportional to the nominal discount rate, whereas it is inversely proportional to the TNPC.

3.5.5 | Economic comparison between the available HRESS in the literature and the suggested one

In the context of the optimally designed HRES for Tumbatu Island, Table 10 provides a comparison with various system configurations previously studied in the literature for other islands by different authors. The comparison is based on the costs associated with each system. While the TNPC value may differ among the proposed HRESs due to variations in initial component costs, average energy demand, and element sizes, the COE value serves as an alternative measure for comparing different combinations of RES.⁹⁴ The table reveals that Hong Kong exhibits a significantly higher COE compared to the other territories, followed by Koh Samui

TABLE 10 Comparison between the available hybrid renewable energy systems in the literature and the suggested one.

Island	Country	Studied energy type	Total net present cost (M\$)	Cost of energy (\$/kWh)	References
RasMusherib	Abu Dhabi Emirates	Photovoltaic (PV)/wind/diesel	14.5	0.2	Rohani and Nour ⁸⁸
KohSamui islands	Thailand	PV/wind/fuel cell/battery	542	0.385	Chaichan et al. ⁸⁹
Deokjeok-do Island	South Korea	PV/wind	11.3	0.159	Back et al. ⁹⁰
Hong Kong	China	PV/wind	0.69	0.595	Ma et al. ⁹¹
Bozcaada island	Turkey	Wind/grid	2.34	0.103	Kalinci and Reviews ⁹²
Masirah Island	Oman	Wind/diesel/natural Gass	129	0.0724	Abdul-Wahab et al. ⁴⁴
Biaro Island	Indonesia	PV/wind	3.87	0.204	Aisyah et al. ⁹³
Kibran Gabriel island	Ethiopia	PV/battery	1.19	0.175	Beza et al. ⁶⁴
Tumbatu Island	Tanzania	PV/wind/grid	4.0	0.09	This study

islands. On the other hand, Masirah and Bozcaada islands have recorded the lowest energy prices in comparison to the mentioned islands. Notably, incorporating a combination of renewable resources and nonrenewable resources, such as diesel and the grid, contributes to decreasing the COE. Conversely, relying solely on renewable systems may result in higher costs, or conversely, depending on the potential and economic parameters of the available renewable sources. Through the results obtained in this techno-economic analysis, satisfactory outcomes have been achieved with low costs compared to the proposed survey, making it an affordable solution for the inhabitants of Tumbatu Island.

4 | CONCLUSION

This study investigated the voltage profile improvement in an HV line at Tumbatu Island as a case study area. The optimization and analysis of HRES comprising solar PV, wind turbine, battery, and the HV line from utility grid was conducted using HOMER software. The findings of this study shed light on various key aspects:

1. The optimization result shows that solar PV and Wind when injected to the grid can produce great improvement of the voltage profile from 29.6 to 31.23 kV on HV side and from 0.219 to 0.23 kV on LV side during maximum demand.
2. Among the configurations analyzed, the HRES incorporating solar PV, wind turbine, and the grid emerged as the most economically viable solution for electricity generation on Tumbatu Island. The integration of RES resulted in a substantial decrease in the

COE, dropping from \$0.114/kWh when relying solely on the grid to \$0.09/kWh. This reduction represents a significant decrease of approximately 21%.

3. The proposed hybrid system has the capacity to meet the growing electricity demand on the island, producing an excess of 104.387 MWh per year.
4. Moreover, the HRES stands out for its environmental friendliness, mitigating techno-economic challenges related to diesel engine emissions and the cost of transporting oil to remote areas.
5. The sensitivity analysis highlighted the impact of solar radiation and wind speed on the COE and TNPC, with higher solar radiation and wind speed resulting in decreased costs.
6. Additionally, the electric load variations were found to affect the solar panel capacity in the system, while the wind turbine capacity remained unaffected. Wind speed was identified as the most sensitive parameter in the study.

In conclusion, the proposed grid-connected HRES improves the voltage profile. With its cost-effectiveness, environmental benefits, and capacity to generate excess electricity, this optimized system presents a promising solution to address the energy needs of the island's inhabitants in a sustainable and eco-friendly manner.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

ORCID

Thani R. Said  <http://orcid.org/0009-0008-7662-561X>

Baraka Kichonge  <http://orcid.org/0000-0002-3998-0910>

Thomas Kivevele  <http://orcid.org/0000-0003-4539-6021>

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APPENDIX A: Inverter specifications

Item	Specifications
Input (DC)	
Max DC power (W)	28,000
Max DC voltage (V)	1000
Min working voltage (V)	250
Output (AC)	
AC nominal power (W)	25,000
Max output current (A)	36
Nominal AC output	50/60 HZ; 400 Vac
Power factor	0.8 leading/lagging
Grid type	3 W/N/PE
Max efficiency (%)	98.6
Safety and protection	
DC reverse-polarity protection	Yes
DC breaker	Yes
Leakage current protection	Yes
Insulation impedance detection	Yes
Residual current protection	Yes